Taylor Series: We define the Taylor series expansion of $f(x)$ about $x=a$ to be the series

$$f(x)=f(a)+f'(a)(x-a)+\frac{f''(a)}{\lfloor 2}(x-a)^2+\cdots+\frac{f^n(a)(x-a)^n}{\lfloor n}+\cdots$$

$$=\sum_{n=0}^{\infty}\frac{f^n(a)(x-a)^n}{\lfloor n}$$

When we truncate this series we obtain the Taylor polynomial of $f(x)$ about $x=a$ of degree n denoted by:

$$P_{n,a}(x)=f(a)+f'(a)(x-a)+\frac{f''(a)}{\lfloor 2}(x-a)^2+\cdots+\frac{f^n(a)}{\lfloor n}(x-a)^n.$$

<u>Example</u>: Find the Taylor series expansion of $f(x)=\cos x$ about $x=\frac{\pi}{2}$.

<u>Solution</u>: Given, $f(x)=\cos x$.

$f\left(\frac{\pi}{2}\right)=0$

$f'(x)=-\sin x$; $f'\left(\frac{\pi}{2}\right)=-1$

$f''(x)=-\cos x$; $f''\left(\frac{\pi}{2}\right)=0$

$f'''(x)=\sin x$; $f'''(\pi/2)=1$

$$\therefore \cos x=f(\pi/2)+f'(\pi/2)(x-\pi/2)+\frac{f''(\pi/2)}{\lfloor 2}(x-\pi/2)^2+\frac{f'''(\pi/2)}{\lfloor 3}(x-\pi/2)^3+\cdots$$

$$=-(x-\pi/2)+\frac{(x-\pi/2)^3}{\lfloor 3}+\cdots$$

Maclaurin Series: We define the Maclaurin series expansion of $f(x)$ about $x=0$ to be the series:

$$f(x)\approx f(0)+f'(0)\frac{x}{\lfloor 1}+f''(0)\frac{x^2}{\lfloor 2}+f'''(0)\frac{x^3}{\lfloor 3}+\cdots$$

$$f(x)\approx\sum_{n=0}^{\infty}\frac{f^n(0)x^n}{\lfloor n}$$

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<u>Example:</u> Find the Maclaurin series expansion of $\frac{1}{1-x}$.

Solution: Given, $f(x)=\frac{1}{1-x}$.

$f(0)=1$. $\quad f'(x)=\frac{1}{(1-x)^2}$ $\quad f'(0)=1$

$f''(x)=\frac{2}{(1-x)^3}$ $\quad f''(0)=2$

$f'''(x)=\frac{6}{(1-x)^4}$ $\quad f'''(0)=6$.

$$\frac{1}{1-x}=f(0)+f'(0)x+f''(0)\frac{x^2}{\lfloor 2}+f'''(0)\frac{x^3}{\lfloor 3}+\cdots$$

$$=1+x+x^2+x^3+\cdots$$

<u>Rolle's Theorem:</u> Let, f be continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b). If $f(a)=0$ and $f(b)=0$, then there is at least one point c in the interval (a,b) such that $f'(c)=0$.

Example: Find the x-intercepts of the function $f(x)=x^2-5x+4$ and confirm that $f'(c)=0$ at some point between the intercepts.

Solution: To find out the x-intercepts we set:

$f(x)=0$

$\Rightarrow x^2-5x+4=0$

$\Rightarrow x^2-4x-x+4=0$

$\therefore x=1,4.$

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Now we set : $f'(x)=0$

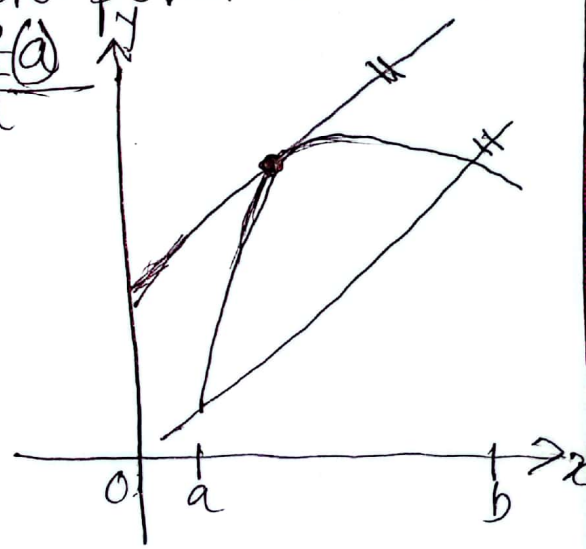
$\Rightarrow 2x-5=0$

$\Rightarrow x=\frac{5}{2}$.

$\therefore f'(x)=0$ at the point $x=\frac{5}{2}$.

<u>Mean Value Theorem:</u> Rolle's theorem is a special case of a more general result called Mean Value Theorem.

Let f be continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b), then there is at least one point c in (a,b) such that : $f'(c)=\frac{f(b)-f(a)}{b-a}$



Geometrically this theorem states that between any two points $(a, f(a))$ and $(b, f(b))$ on the graph of a differentiable function f, there is at least one point where the tangent line to the graph is parallel to the secant line.

Example: Show that the function $f(x)=\frac{1}{4}x^3+1$. satisfies the hypotheses of the Mean Value Theorem over the interval $[0,2]$ and find all values of c in the interval $(0,2)$.

Solution: Given, $f(x)=\frac{1}{4}x^3+1$

$$f(0)=0+1=1$$

$$f(2)=\frac{1}{4}\times 2^3+1=3.$$

We set, $f'(x)=\frac{3}{4}x^2$.

and $\frac{f(b)-f(a)}{b-a}=\frac{f(2)-f(0)}{2-0}=\frac{3-1}{2-0}=\frac{2}{2}=1$

According to the hypotheses of Mean Value Theorem

$$f'(x)=1$$

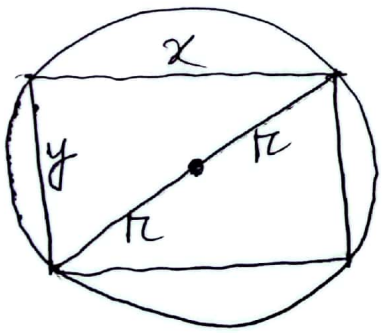
$$\Rightarrow \frac{3}{4}x^2=1$$

$$\therefore x=\pm\frac{2}{\sqrt{3}}$$

$$\therefore x=\frac{2}{\sqrt{3}}.$$

Optimization Problem:

Example: Find the dimension of the rectangle with maximum area that can be inscribed in a circle of radius 10 cm.



Solution: Let,

r = radius of the circle.

x = length of the rectangle.

y = width of the rectangle.

A = area of the rectangle.

Here, $A = xy$ —— (1)

From the figure: $x^2 + y^2 = (2r)^2$

$$\Rightarrow x^2 + y^2 = 4r^2$$

$$\Rightarrow y^2 = 4r^2 - x^2$$

$$\therefore y = \sqrt{4r^2 - x^2}$$

We need to maximize A so that it can be inscribed in the circle of radius 10 cm. Also, we know that $0 \leq x, y \leq 20$. Differentiating A with respect to x:

$$\frac{dA}{dx} = \sqrt{400 - x^2} - \frac{x^2}{\sqrt{400 - x^2}}$$

Setting $\frac{dA}{dx} = 0$

$$\Rightarrow \sqrt{400 - x^2} - \frac{x^2}{\sqrt{400 - x^2}} = 0$$

$$\Rightarrow \frac{400 - x^2 - x^2}{\sqrt{400 - x^2}} = 0$$

$$\Rightarrow \frac{400 - 2x^2}{\sqrt{400 - x^2}} = 0$$

$$\Rightarrow 2x^2 = 400$$

$$\Rightarrow x^2 = 200$$

$$\therefore x = 10\sqrt{2}.$$

$$\frac{d^2A}{dx^2} = \frac{\sqrt{400 - x^2}(-4x) - (400 - 2x^2)\frac{(-2x)}{2\sqrt{400 - x^2}}}{400 - x^2}$$

$$= \frac{-4x\sqrt{400 - x^2} + x\frac{(400 - 2x^2)}{\sqrt{400 - x^2}}}{400 - x^2}$$

< 0 at $x = 10\sqrt{2}$

$\therefore$ The dimensions of the rectangle are $10\sqrt{2}$ cm and $10\sqrt{2}$ cm.

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Example: A liquid form of antibiotic manufactured by a pharmaceutical firm is sold in bulk at a price of \$200 per unit. If the total production cost (in dollars) for x unit is: $c(x) = 500000 + 80x + 0.03x^2$ and if the production capacity of the firm is at most 30,000 units in a specified time, how many units of antibiotic must be manufactured and sold in that time to maximize the profit?

<u>Solution:</u> Let the amount need to be produced be x units.

Total revenue, $R(x) = 200x$.

Profit = Total revenue − Total cost.

$\Rightarrow P(x) = R(x) - c(x)$

$\Rightarrow P(x) = 200x - 500000 - 80x - 0.03x^2$

Setting $\frac{dP}{dx} = 0$

$\Rightarrow 120 - 0.06x = 0$

$\therefore x = 2000$

Here, $\frac{d^2P}{dx^2} = -0.06 < 0$

Since, the second derivative is greater than for all values of x, the profit is maximized at $x = 2,000$ units.

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Example: Find a point on the curve $y=x^2$ that is closest to the point $(18,0)$.

Solution: let any arbitrary point on the curve be (x,y). The distance L between $(18,0)$ and (x,y):

$$L=\sqrt{(x-18)^2+(y-0)^2}=\sqrt{(x-18)^2+y^2}.$$

Since (x,y) lies on the curve; x and y satisfy $y=x^2$; thus: $L=\sqrt{(x-18)^2+x^4}$

The distance L and the square of the distance L^2 are minimized at the same value. Thus, the minimum value of $S=L^2=(x-18)^2+x^4$ occur at the same x-value.

$$\frac{dS}{dx}=2(x-18)+4x^3=4x^3+2x-36.$$

So the critical points satisfy:

$$4x^3+2x-36=0$$

$$\Rightarrow 2x^3+x-18=0$$

$\Rightarrow x=2$ is the only real solution.

To determine the measure of this critical point, we will use the second derivative test:

$$\frac{d^2S}{dx^2}=12x^2+2, \text{ so } \left.\frac{d^2S}{dx^2}\right|_{x=2}=50>0.$$

which show that a relative minimum occurs at $x=2$. Thus the point of the curve $y=x^2$ closest to $(18,0)$ is: $(x,y)=(x,x^2)=(2,4)$ Ans

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Optimization Problem

① A garden is said to be laid out in a rectangular area and protected by a chicken wire fence. What is the largest possible area of the garden if only 100 running feet of chicken wire is available for the fence?

② Find a point on the curve $y = x^2$ that is closest to a point $(18, 0)$.

③ A chemical manufacturer sells ~~sulp~~ sulfuric acid in bulk at a price of \$100 per unit. If the daily total production cost in dollars for x unit is: $C(x) = 1,00,000 + 50x + 0.0025x^2$ and if the daily production capacity is at most 4,000 units, how many units of sulfuric acid must be manufactured and sold daily to maximize the profit?

④ A fertilizer producer finds that it can sell its product at a price of $p = 300 - 0.1x$ dollars per unit when it produces x units of fertilizer. The total production cost (in dollars) for x units is: $\boxed{C(x) = 15000 + 125x + 0.025x^2}$ and if the daily production capacity is at most 4000 units, how many units of sulfuric acid must be manufactured and

and sold daily to maximize the profit?

④ A fertilizer producer finds that it can sell its product at a price of $P = 300 - 0.1x$ dollars per unit when it produces x units of fertilizers. The total production cost (in dollars) for x unit is:

⑤ A firm determines that x units of its product can be sold daily at P dollars per unit where: $x = 1000 - P$

The cost of producing x units per day is:

$$c(x) = 3000 + 20x.$$

(a) Find the revenue function $R(x)$.

(b) Find the profit function $P(x)$.

(c) Assuming that the production capacity is at most 500 units per day, determine how many units the company must produce and sell each day to maximize the profit.

(d) Find the maximum profit.

(e) What price per unit must be changed to obtain the maximum profit?

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→ A function f of two variables, x and y, is a rule that assigns a unique real number $f(x,y)$ to each point (x,y) in some set D in the xy-plane.

→ A function f of three variables, x, y, and z is a rule that assigns a unique real number $f(x,y,z)$ to each point (x,y,z) in some set D in the three-dimensional space.

<u>Example:</u> Let, $f(x,y) = \sqrt{y+1} + \ln(x^2-y)$. Find

(i) $f(e,0)$ (ii) $f(x,0)$ (iii) $f(e,y)$.

<u>Solution:</u> Given: $f(x,y) = \sqrt{y+1} + \ln(x^2-y)$

$$f(e,0) = \sqrt{0+1} + \ln(e^2-0) = 1 + \ln e^2$$
$$= 1 + 2\ln e = 1+2 = 3$$
$$f(x,0) = \sqrt{0+1} + \ln(x^2-0) = 1 + 2\ln(x)$$
$$f(e,y) = \sqrt{y+1} + \ln(e^2-y).$$

<u>Problem:</u> Find the domain of the function:

$$f(x,y) = xe^{(-\sqrt{y+2})}$$

<u>Solution:</u> Given, $f(x,y) = xe^{(-\sqrt{y+2})}$

There is no value of x for which the function $f(x)$ is undefined, so $x \in (-\infty, \infty)$. For y, we set:

$y+2 \geq 0$

$\Rightarrow y \geq -2$.

$\therefore y \in [-2, +\infty)$

So, the domain of the function is all points above or on the line $y = -2$.